Microsoft

Azure Cosmos DB Closing Session (latest Features and Q&A)

The universe’s most **scalable** database for AI apps

performant

cost-effective

flexible

Azure Cosmos DB
For NoSQL

Serverless or Provisioned Throughput

Automatic data partitioning

Instant and dynamic autoscale

Low latency, real-time data transactions

Mission critical reliability (99.999%)

Built-in vector indexing and search w/ DiskANN

A Serverless Database for Any Application

Azure Serverless App Ecosystem

Cost efficient & flexible

Start simply

Pay only for what you use

Low cost: $0.25 USD / 1M RUs

Easy to configure & manage

Great for multitenant apps, MVPs and development

AI Use Cases with Azure Cosmos DB

Operational + Vector Database

No ETL

Consistent data

Reduce complexity & costs

Retrieval Augmented Generation (RAG)

Personalize LLM on your data

Cheaper than fine tuning

Faster iteration on new data

Conversational History

Conversational context

UX improvements

LLM optimizations

Auditing

Semantic Caching

Drastically reduces latency

Saves on Token consumption

Reduces costs and latency for LLM

AI Agents

Agent-to-agent transactions

Agentic Memories

State Events Stores

Agents Logging

Non-relational Databases on Azure

Why do customers choose NoSQL DB

95%+ of all customers have non-relational Databases, and are growing their usage to keep the pace of AI Innovation

Best resiliency, scalability, and performance for agentic apps

Azure Cosmos DB
(NoSQL API)

Optimized for Scale Out

Industry leading high availability (5x 9's)

Performance at any scale

Fastest Autoscale + Serverless

Build MongoDB-compatible apps Open Source, Hybrid/Multi-Cloud

Azure DocumentDB
(formerly vCore-based Azure Cosmos DB for MongoDB API)

Optimized for Scale Up

Familiar vCore model + MongoDB API

Open Source, Hybrid/Multi-Cloud

Rich query and transactions

What’s new for Azure Cosmos DB @ BUILD

Generally Available

- Global Secondary Index
- Per-Partition Automatic Failover
- Change partition keys
- All versions and deletes change feed mode
- Azure Cosmos DB Linux Emulator
- MCP Toolkit for Azure Cosmos DB
- Agent Kit for Azure Cosmos DB

Public preview

- Distributed Transactions
- Azure Backup for Azure Cosmos DB
- AI assistance in Azure Cosmos DB extension for VS Code
- Semantic Reranking
- Integrated Embeddings
- Agent Memory Toolkit
- Agentic Retrieval Toolkit
- OmniVec Toolkit

What you have learned today?

Azure Cosmos DB Fundamentals
- Environment setup, provisioning accounts, configuring indexing and partitioning, and SDK installation.
- Data modelling, partitioning strategies, schema design, and query tuning.
- Writing and optimising SQL queries, analysing RU consumption, and diagnosing performance bottlenecks.

AI Integration
- Implementing AI Agents with Cosmos DB, including full text, vector, and hybrid search.
- Building AI-powered search solutions and evaluating data modelling techniques for AI workloads.

Security & Cost Optimisation:
- Role-based access control (RBAC), private endpoints, encryption.
- Cost-saving strategies: autoscale, TTL, indexing tuning, and live optimisation in lab exercises.

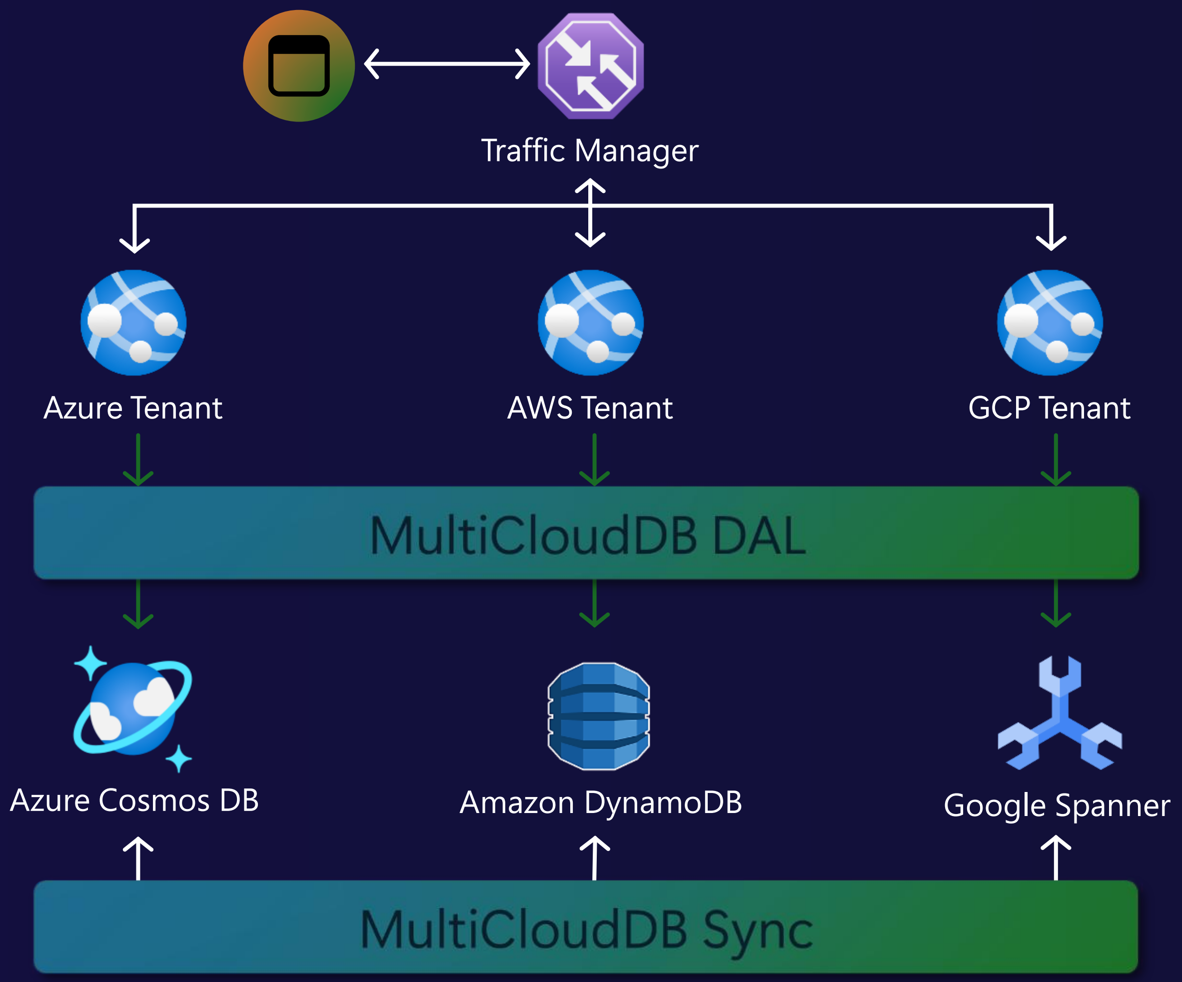
Latest Innovations:
- Recent feature releases: burst capacity, hierarchical partitioning, integrated vector indexing.
- Recap of key learnings and open Q&A.

Thank you for participating!

You are now equipped to build scalable, high-performance, and cost-optimized applications with Azure Cosmos DB

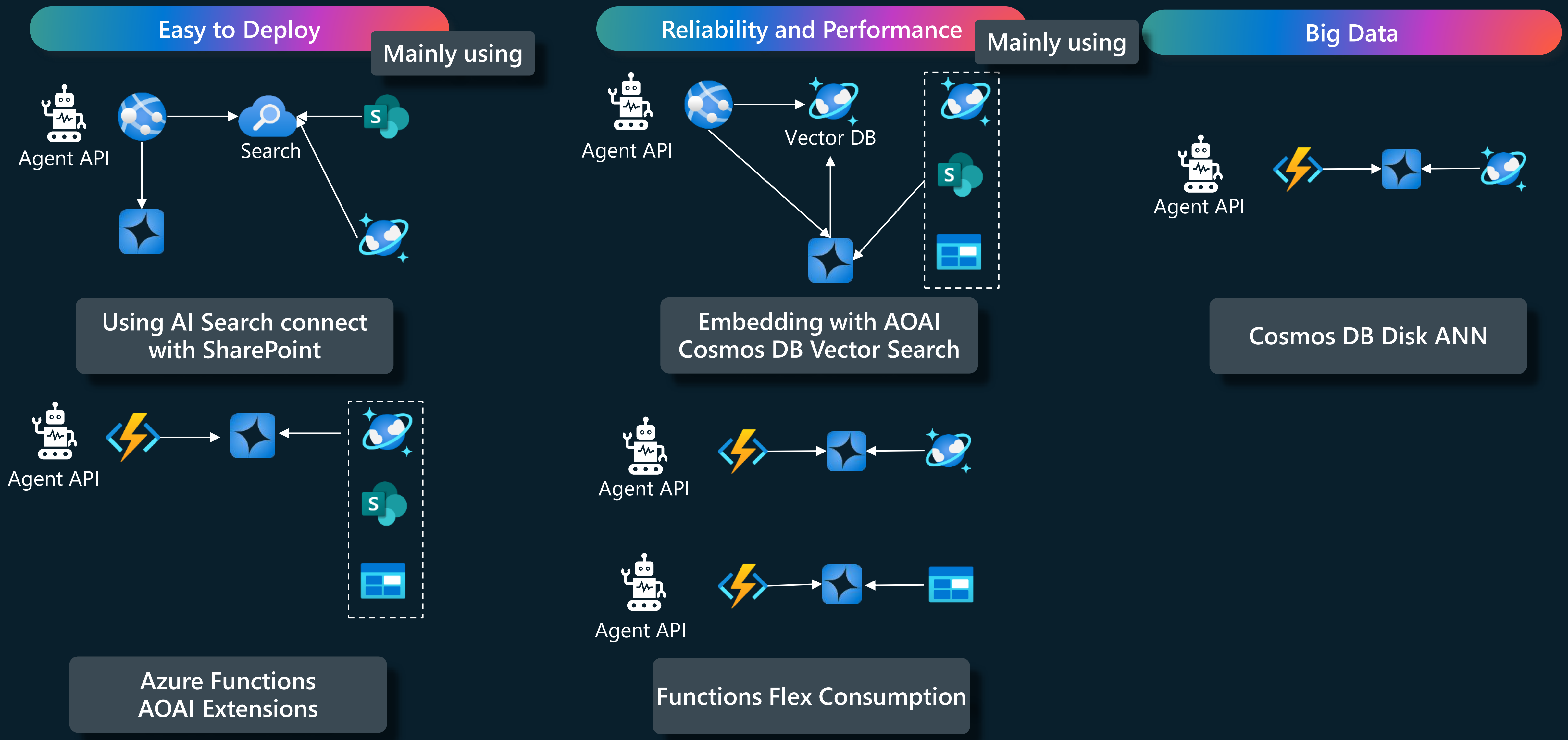
MultiCloudDB: Write Once. Run Anywhere.

Cross-cloud portability maintaining 5'9s SLA

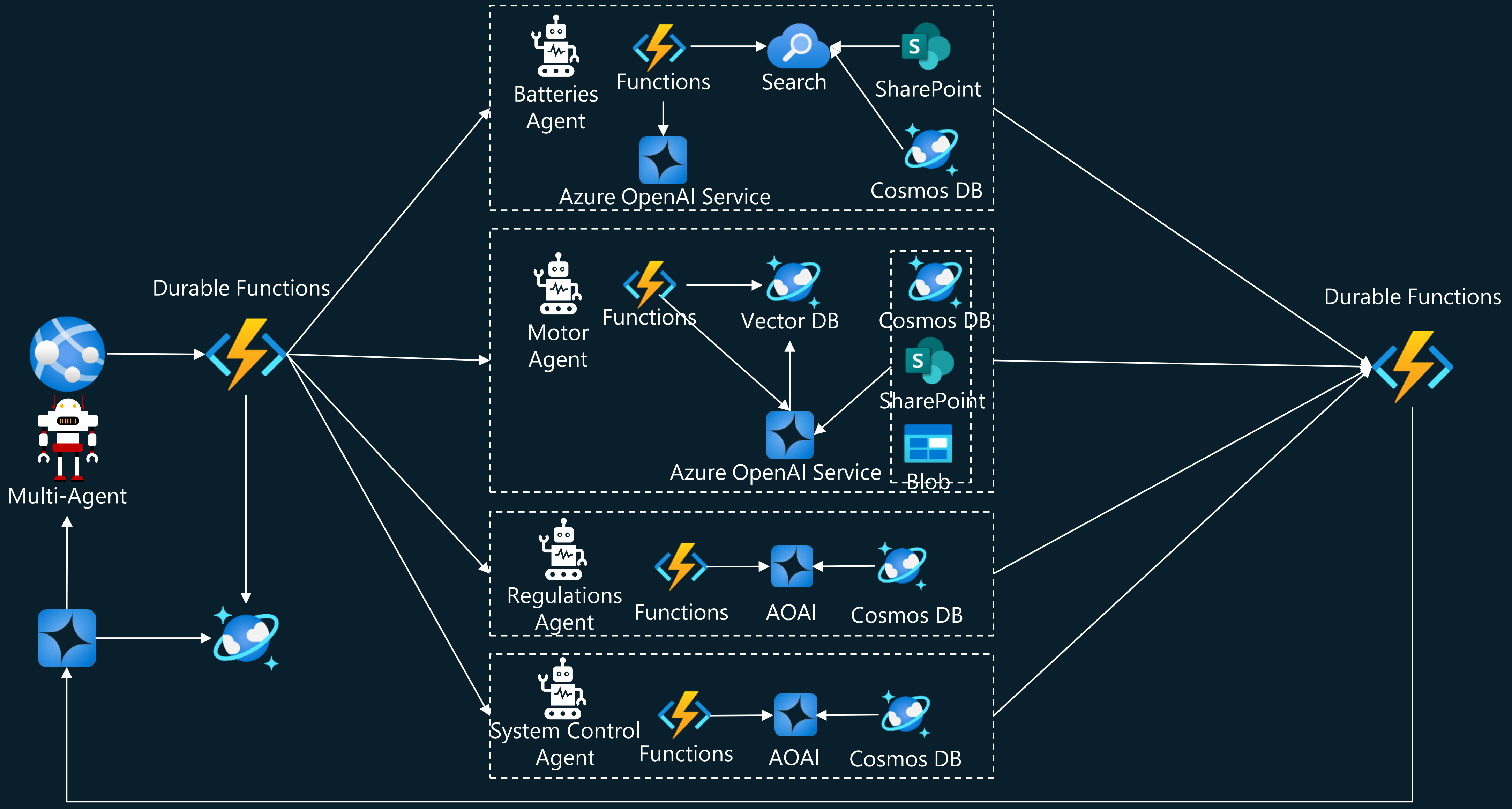


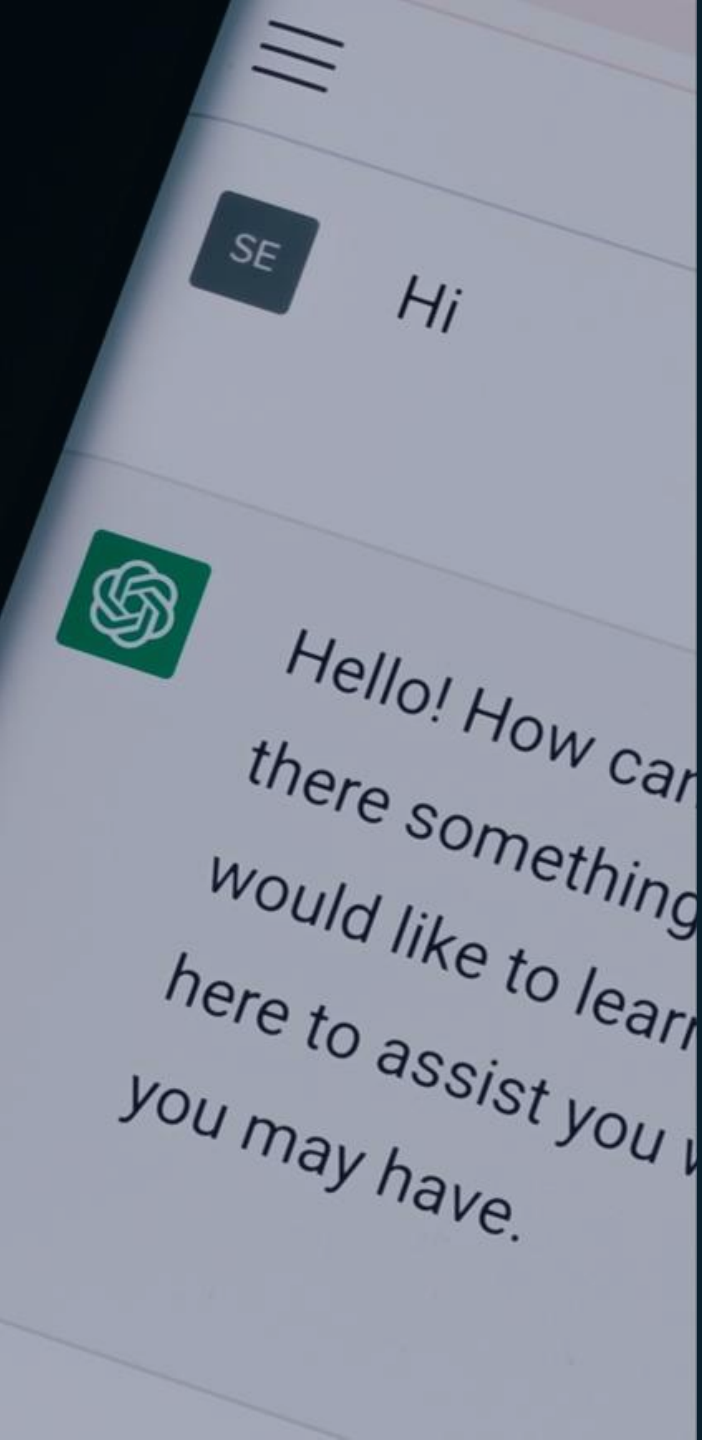
- Portability across 3 clouds
- Cross-cloud replication
- Cross-cloud DR
- Open Source, MIT Licensed
- Aka.ms/multiclouddb

TOYOTA Multi-Agent RAG Pattern



Multi-Agent using Durable Functions





ChatGPT scales with Azure Cosmos DB

Open AI stores ChatGPT conversations and all other user interactions in Azure Cosmos DB, 40+ workloads

OpenAI

Challenge

Meet incredible demand from growth and traffic spikes, without having to worry about database operations

Outcomes

- Rapidly and seamlessly scaled as service grew, with zero downtime
- Able to iterate fast on data shapes thanks to schemaless flexibility
- Consistent high performance and availability

Key Azure products used:

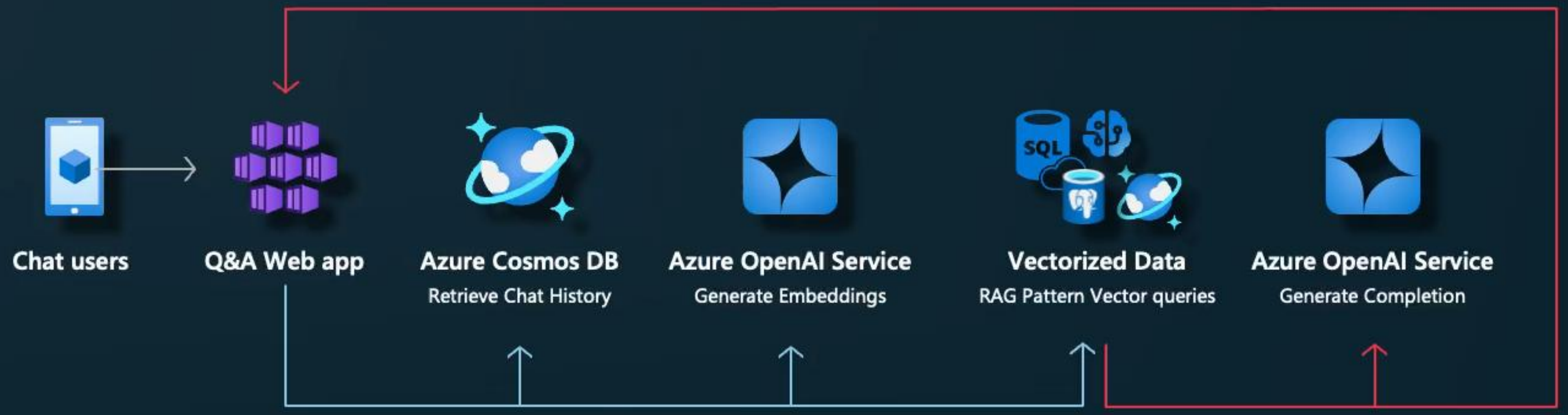
Azure Cosmos DB

Azure Kubernetes Service

Azure AI Search

No Semantic Cache

- Every single request, even same questions, completes this path.
- Extremely high cost and latency for large payloads from RAG Pattern Data.



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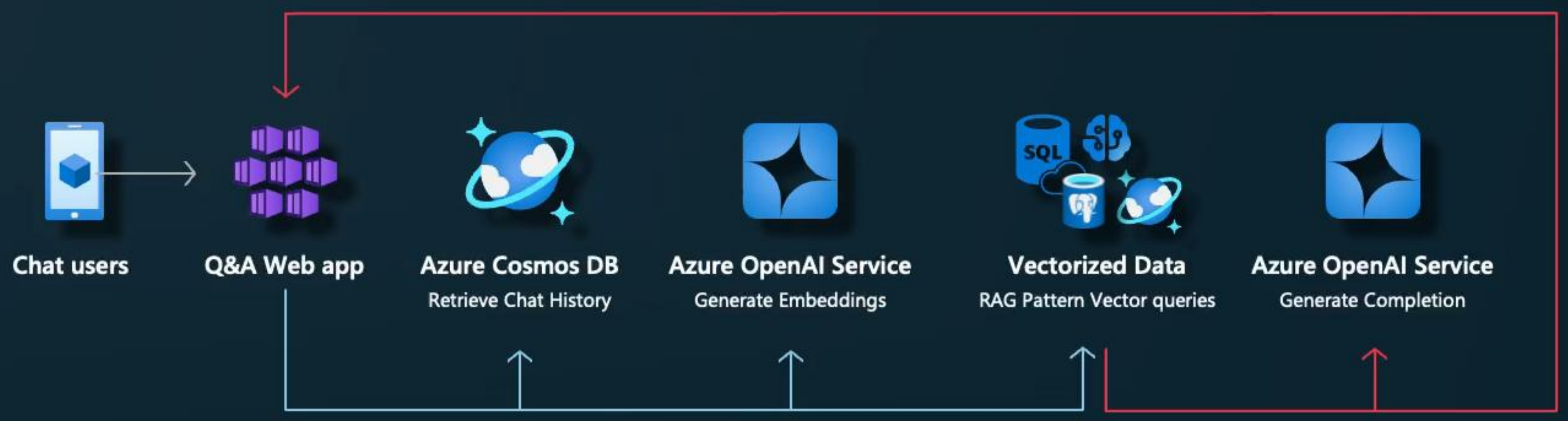

With Semantic Cache

- Repeated questions handled by cache. Only unique requests completes full path.
- Zero token cost for LLMs. Extremely fast.